USER MANAGEMENT SYSTEM

1. Title

Industry-Oriented Full Stack User Management System

---

2. Objective

The objective of this project is to design and develop a full stack web application that provides secure user registration, authentication, and role-based access control using MERN/MEAN stack technologies.

---

3. Problem Statement

Most business applications require secure systems to manage users, authentication, and access control. Poorly designed user management systems can lead to security vulnerabilities and poor user experience. This project aims to build a scalable and secure solution for managing users and roles effectively.

---

4. Technologies Used

- Frontend: React.js / Angular
- Backend: Node.js, Express.js
- Database: MongoDB
- Authentication: JSON Web Token
- Password Security: bcryptjs
- API Type: REST APIs
- Tools: Git, Postman

---

5. System Architecture

The application follows a client-server architecture:

- The frontend (client) interacts with the backend through REST APIs
- The backend processes requests and performs business logic

- The database stores user information securely
- Authentication is handled using tokens

Flow:
User → Frontend → Backend → Database → Backend → Frontend

---

6. Features Implemented

User Features

- User registration
- User login
- View and update user profile

Admin Features

- View all registered users
- Role-based access control

---

7. Functional Requirements

- User should be able to register with valid details
- User should be able to login with correct credentials
- System should generate a token after successful login
- Only authenticated users should access protected routes
- Admin should have access to user management features

---

8. Non-Functional Requirements

- Security of user data
- Fast response time
- Scalability of backend services
- User-friendly interface

---

9. API Endpoints

- POST /api/register – Register a new user
- POST /api/login – Login user and generate token
- GET /api/profile – Get logged-in user details
- GET /api/users – Get all users (admin only)

---

10. Database Design

User Collection

Fields:

- name
- email
- password (hashed)
- role (admin/user)
- createdAt

---

11. Security Features

- Password hashing using bcrypt
- Token-based authentication using JWT
- Protected routes for authorized access
- Role-based authorization

---

12. Input Validation and Error Handling

- Validation for email and password fields
- Error messages for invalid login credentials
- Handling of duplicate user registration
- Server-side validation for all inputs

---

13. Testing

Testing was performed using Postman to verify:

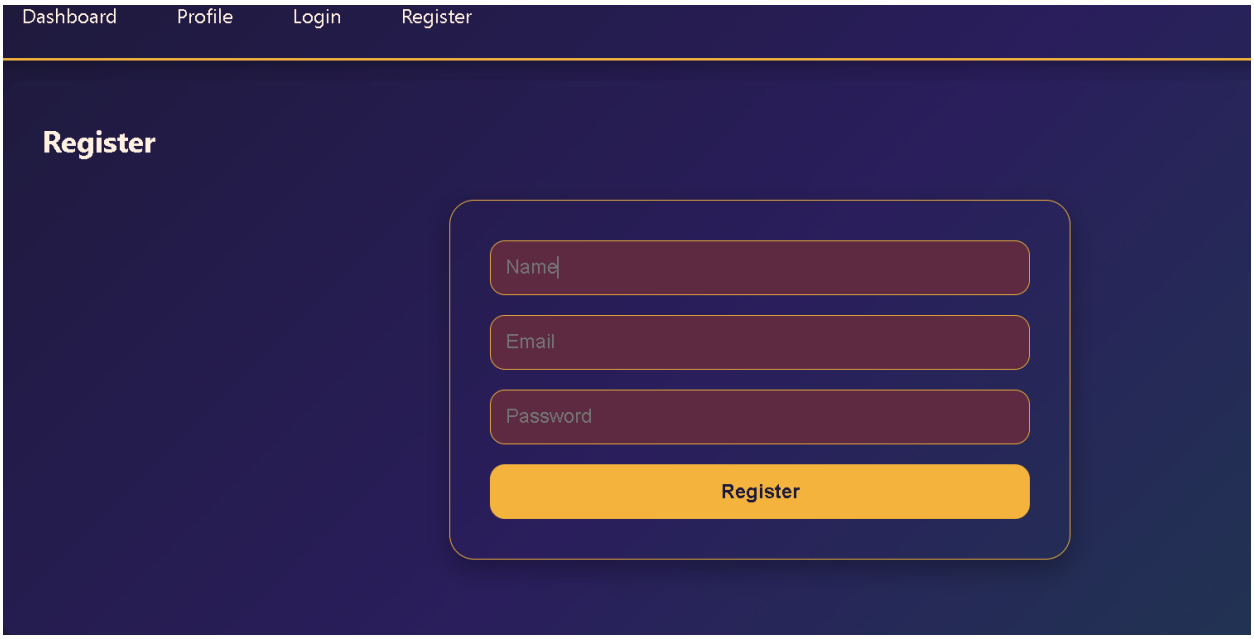
- User registration

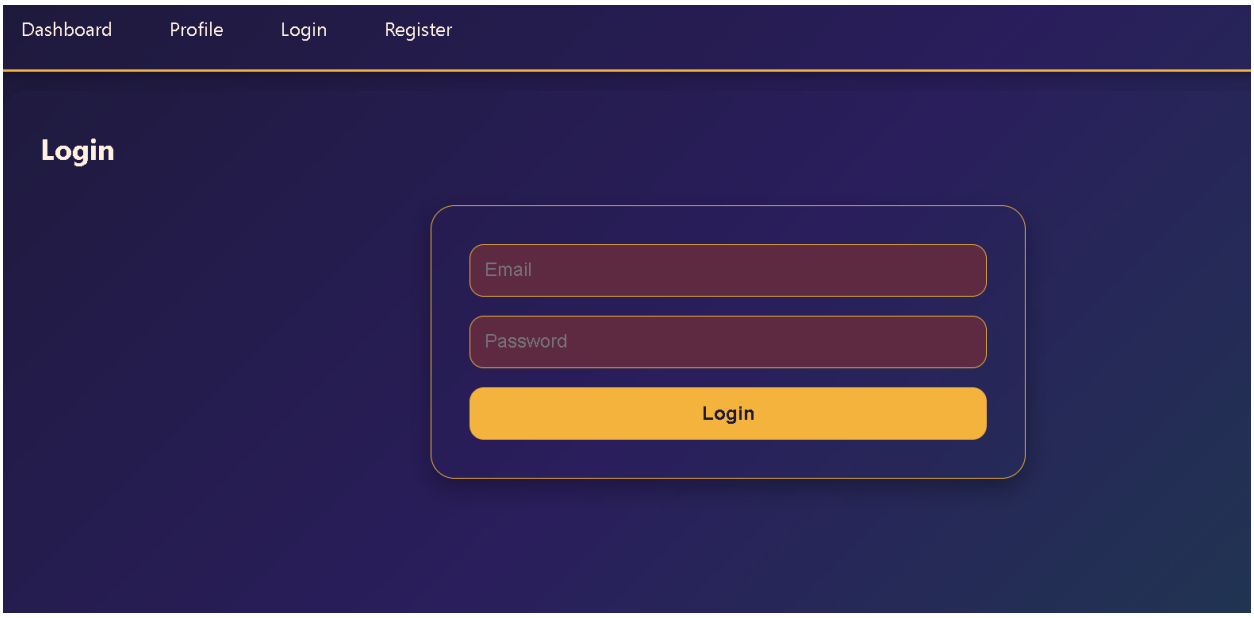
- Login functionality
- Token authentication
- Access control

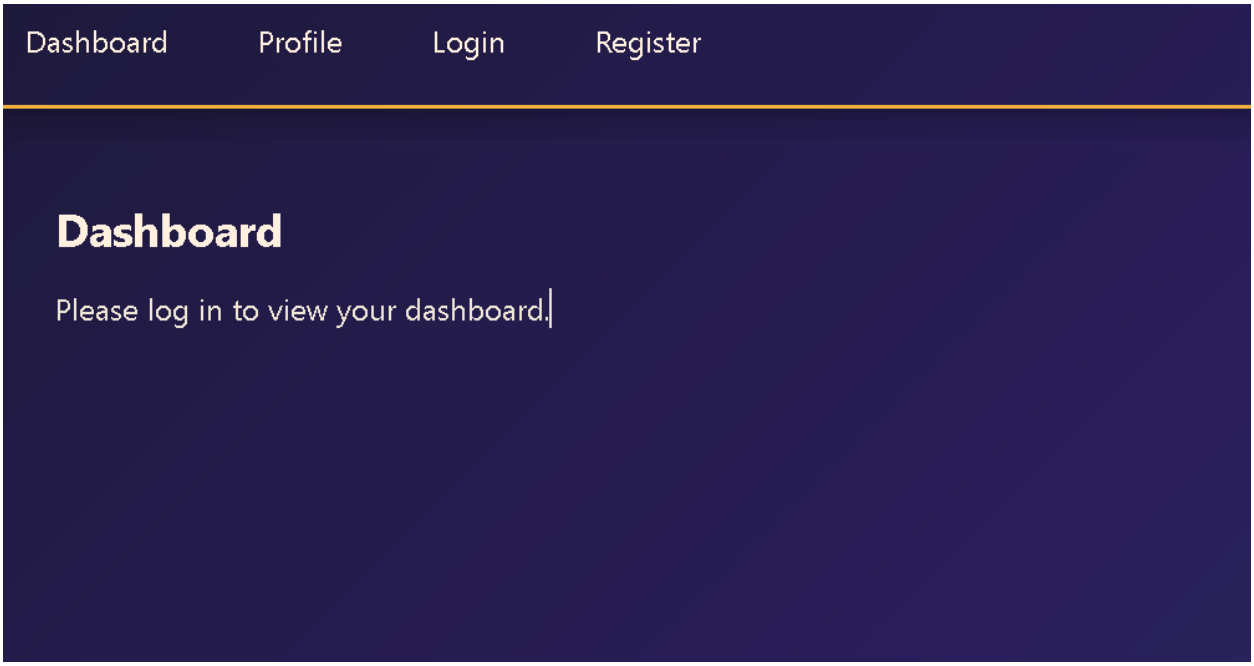
---

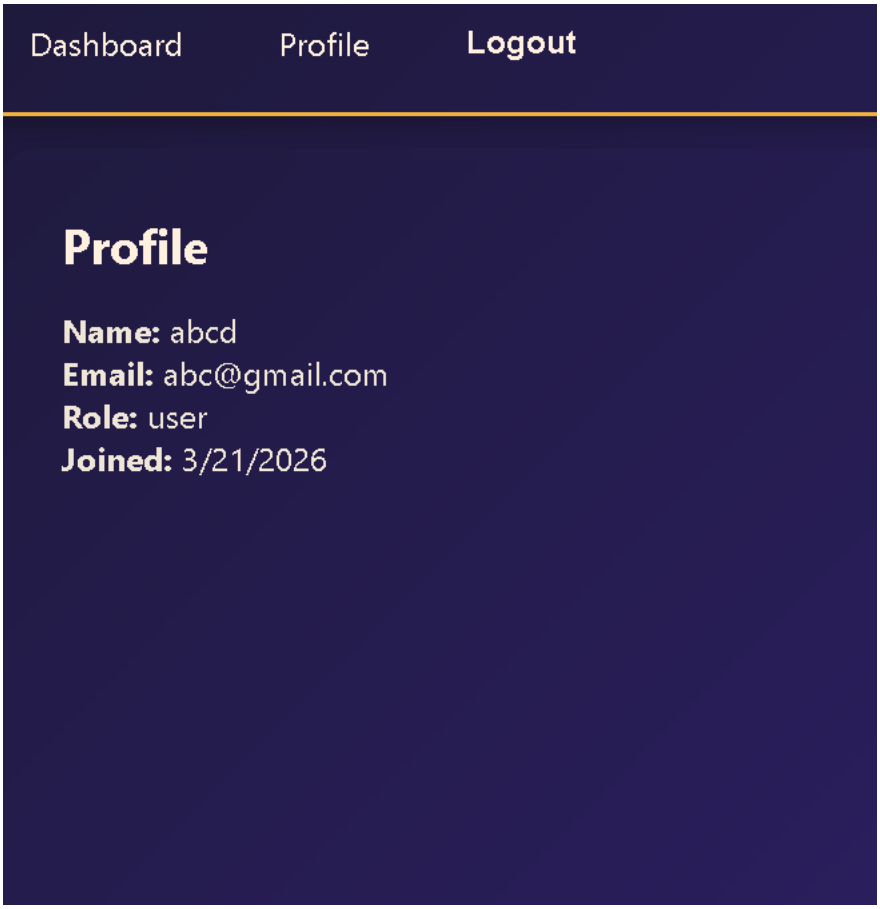
14. Output

- Registration page
- Login page
- Dashboard
- Profile page




Dashboard
Profile
Login
Register
Login
Email
Password
Login


Dashboard
Profile
Login
Register
Dashboard
Please log in to view your dashboard.



---

15. Deployment

- Frontend deployed using Vercel or Netlify
- Backend deployed using Render or Railway
- Database hosted on MongoDB Atlas

---

16. Conclusion

The User Management System successfully demonstrates a full stack application with secure authentication and role-based access control. It provides practical experience in integrating frontend and backend technologies and implementing real-world security features.